

Karam Veer Singh

Semiconductor Process Development | Thin-Film Processing | Vacuum Processing & Equipment Integration

PhD Candidate (Thesis Submitted) & Project Scientist — Indian Institute of Technology Bombay

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PROFESSIONAL SUMMARY

PhD candidate in Materials Science at IIT Bombay specializing in semiconductor thin-film process development, vacuum processing, and semiconductor equipment integration. Experienced in developing CVD- and PVD-based thin-film processes, commissioning and qualifying semiconductor processing equipment, troubleshooting process and hardware issues, executing process qualification experiments, and translating laboratory-scale process development into reproducible manufacturing processes. Currently extending this expertise as a Project Scientist on a National Quantum Mission project, building a new MOCVD deposition platform at IIT Bombay.

CORE COMPETENCIES

Semiconductor Process Development • Thin-Film Deposition (CVD/PVD) • Equipment Integration & Commissioning • Process Qualification & Validation • Process Optimization & Troubleshooting • Vacuum Systems • Root Cause Analysis • Process-Structure-Property Relationships • Surface & Interface Engineering • Semiconductor Device Fabrication

TECHNICAL SKILLS

Thin-Film Deposition: Chemical Vapor Deposition (Hot-Wire CVD), RF Magnetron Sputtering, Thermal Evaporation, Electron-Beam Evaporation (Teaching Assistant), DC Magnetron Sputtering (Teaching Assistant)

Semiconductor Processing Equipment & Vacuum Systems: Multi-chamber vacuum processing systems, cluster tool integration & commissioning, equipment qualification, load-lock systems, vacuum pumps & gauges, leak detection, mass flow controllers (MFCs), gas delivery systems, pressure calibration

Characterization: Thickness Profilometry, FTIR, Raman, & UV-Visible Spectroscopy, Hall Effect Measurement, Minority Carrier Lifetime (QSSPC), External Quantum Efficiency (EQE), Current-Voltage (I-V) Measurement

Process Engineering & Development: Semiconductor process development, thin-film process optimization, process troubleshooting, root cause analysis, surface & interface engineering, thin-film growth kinetics, process-structure-property correlation, systematic experimental design, process data analysis & interpretation

Process Safety: Handling of pyrophoric and toxic process gases, HF-based wet processing

Software: Origin, CorelDRAW, C Programming

EDUCATION

PhD Candidate (Thesis Submitted; Defense Pending) — Materials Science

Supervisor: Prof. Rajiv O. Dusane, Department of Metallurgical Engineering & Materials Science (MEMS), Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India

Thesis: Technology Development of Silicon Heterojunction Solar Cells Through Hot-Wire Chemical Vapor Deposition: From Interface Engineering to Vacuum-Integrated Processing

CGPA: 8.88/10

Master of Technology (M.Tech) — Energy Technology

School of Engineering, Tezpur University, Assam, India | 2015–2017

Research Focus: Luminescent Solar Concentrators (LSCs) for Building-Integrated Photovoltaics (BIPV)

Bachelor of Technology (B.Tech) — Electrical Engineering

Assam Engineering College, Guwahati, India | 2009–2013

CURRENT POSITION

Project Scientist, National Quantum Mission (NQM) — Electronic Materials Processing Laboratory (EMPL)

PI: Prof. Tanushree Choudhury, MEMS, Indian Institute of Technology Bombay, Mumbai, India | Aug 2026 – Present

Present Working on wafer-scale semiconductor emitters and detectors, with EMPL's scope focused on 2D material growth and characterization for device fabrication.

- Validating technical specifications and building out the component list for a new MOCVD deposition system.
- Sourcing vendors and preparing quotation requests for remaining tool components under project funding constraints.
- Contributing to the procurement and build-out of a new MOCVD setup, from specification through equipment integration.

SEMICONDUCTOR PROCESS ENGINEERING EXPERIENCE

Thin-Film Process Development

- Developed and optimized HWCVD amorphous silicon thin-film deposition processes for crystalline silicon surface passivation, achieving minority carrier lifetimes approaching 1 ms.
- Established process-structure-property relationships by correlating deposition conditions with thin-film microstructure, material properties, and electrical performance.
- Optimized HWCVD-grown doped a-Si:H carrier-selective layers, improving electrical conductivity while investigating microstructural evolution.
- Developed HWCVD-grown a-SiN:H anti-reflection coatings and integrated passivation stacks, achieving <1% optical reflectance.
- Deposited RF magnetron sputtered indium tin oxide (ITO) transparent conductive oxide films for silicon heterojunction device fabrication.
- Fabricated and electrically characterized silicon heterojunction (SHJ) solar cells to validate developed thin-film processes.

Process Equipment Integration & Commissioning

- Independently led the integration, commissioning, and qualification of a multi-chamber semiconductor processing platform incorporating HWCVD, RF magnetron sputtering, load-lock, and vacuum-integrated wafer transfer.
- Defined process, equipment, and operational requirements for equipment functionality, including wafer handling, process sequencing, automation workflows, and safety interlocks, in collaboration with equipment vendors.
- Integrated and qualified vacuum, process gas delivery, cooling, exhaust, and utility systems to enable reliable operation of the complete processing platform.
- Performed systematic troubleshooting and root-cause analysis of mechanical, vacuum, gas handling, thermal, electrical, and process-related issues to improve equipment reliability, process stability, and repeatability.
- Executed process qualification through thin-film deposition, characterization, and iterative hardware-process optimization to establish baseline process capability.

Semiconductor Equipment Operations & Troubleshooting

- Operated, maintained, qualified, and troubleshot semiconductor thin-film deposition systems, diagnosing and resolving vacuum, mechanical, electrical, gas delivery, and process-related equipment issues to maximize equipment reliability and experimental uptime.
- Performed preventive and corrective maintenance, including reactor assembly/disassembly, leak detection, vacuum qualification, equipment cleaning, and component replacement to ensure reliable process operation.
- Served as Teaching Assistant for graduate laboratory courses, demonstrating vacuum technology, DC/RF magnetron sputtering, electron-beam evaporation, and semiconductor thin-film deposition principles.

INDUSTRIAL & INTERNATIONAL RESEARCH COLLABORATION

Applied Materials India — Industrial Process Development

- Contributed to an industrial process development study on copper silicidation by executing vacuum thermal processing experiments and collaborating with Applied Materials engineers to evaluate process windows and identify process and equipment limitations.

University of York, UK — Thin-Film Radiation Detector Development

- Supported the fabrication of HWCVD-grown amorphous silicon radiation detector prototypes by contributing to process development, thin-film deposition, device fabrication, and mentoring junior researchers in safe semiconductor processing practices.

PUBLICATIONS

- Microstructure-Dependent Response of HWCVD a-Si:H Passivation Layers to Subsequent a-SiN:H Deposition: Implications for Optical and Surface Passivation — Under Review
- Pulsed Hot-Wire CVD of a-Si:H Films: From Catalytic SiH₄ Dissociation Model to Experiments — Under Review

AWARDS

- Best Poster Award – Metals & Materials Summit, IIT Bombay (2022)
- Best Oral Presentation – Recent Advances and Innovation in Solar Energy, IIT Madras (2021)
- Best Oral Presentation – Annual Department Symposium, IIT Bombay (2019)

PROFESSIONAL DEVELOPMENT

Conferences

- International Conference on Crystalline Silicon Photovoltaics (SiliconPV) – TU Delft, Netherlands (2023)
- International Conference on Hot-Wire (Catalytic) Chemical Vapor Deposition (HWCVD) (Online) (2023)
- International Conference on Photovoltaics, Solar Thermal, Energy Storage and Solar Fuels (Online) (2021)
- International Symposium on Metastable, Amorphous and Nanostructured Materials (ISMANAM) – Chennai, India (2019)

Workshops

- Next-Generation Photovoltaics (nPV) Workshop – TU Delft, Netherlands (2023)
- Silicon-Based Radiation Detectors and Applications Workshop – IIT Bombay & University of York (2024)